NidhiRoute AI

Smart Scheme Discovery & Application Guidance Platform

PROJECT REPORT | Academic / Hackathon Prototype

Project purpose: A guided platform that helps SC students and entrepreneurs explore configured funding schemes, complete a basic eligibility screen, estimate loan repayments, find demo channel partners and prepare supporting documents.

Project status: Prototype for academic demonstration. Scheme data, partner information and eligibility settings require official verification before real-world use.

1. Executive Summary

NidhiRoute AI is a web-based prototype designed to simplify the early stages of scheme discovery and application preparation. It combines a searchable scheme catalogue, a general scheme-matching journey, dedicated application pages for selected schemes, configured eligibility checks, EMI estimation, map-based partner discovery and document screening. The interface guides applicants from the home page and login through dashboard, scheme application and document submission.

The system is intended to improve access to information and reduce confusion during application preparation. It does not make official eligibility decisions, approve loans or authenticate identity documents. Final decisions remain with the relevant scheme authority, bank or institution.

2. Problem Statement

Applicants may find it difficult to identify schemes suited to their purpose, understand basic configured eligibility conditions, estimate repayment commitments and determine which supporting documents to prepare. Scheme information and application steps can be spread across different sources.

3. Objectives

- Provide a single interface to browse and search education and business funding schemes.
- Guide users through login, dashboard and scheme-specific application pages.
- Apply transparent prototype eligibility rules and explain the screening result.
- Estimate EMI and repayment information where the selected product is a loan.
- Display configured channel-partner locations and external map directions.
- Collect required documents through file upload or camera capture and perform preliminary OCR-based screening.

4. Scope and Main Modules

Home and account access

The landing page introduces NidhiRoute AI. Applicants can sign in or create an account before proceeding to the dashboard. The backend stores password hashes using Node.js scrypt rather than plaintext passwords.

Applicant dashboard and scheme catalogue

The dashboard provides scheme search and purpose/type filters. Each scheme card can open a dedicated application page for that selected scheme. The general matching page remains available for users who want to compare recommendations.

General scheme matching

The general journey collects applicant profile information and presents configured eligibility analysis, explainable scheme recommendations, scheme comparison, EMI estimation, partner routing and tracking elements.

Dedicated scheme application

A scheme card opens a page for that scheme only. The purpose is preselected. The form collects applicant name, annual family income, project/course cost, location and preferred timeline. After screening, users can review partner locations and continue directly to document verification.

Document verification and submission

The document page supports PDF and image uploads and camera capture. PDF.js reads PDF text and renders pages for OCR; Tesseract.js performs OCR; jsQR is used for QR-code detection. Document-specific keyword checks and applicant-name checks are preliminary screening only. The backend saves files locally when run on the applicant's computer.

5. Configured Prototype Rules

These rules reflect the current project configuration and must not be interpreted as universal or official government eligibility rules:

Rule	Configured behavior
Annual Family Income	AFI must be strictly below ₹5,00,000 for every scheme. ₹5,00,000 or above fails the configured income threshold.
Loan amount	Loan applications are capped at ₹5,00,000, subject to a lower maximum configured for the selected scheme.
Education loan additional document	For an education-loan application with AFI below ₹1,00,000, Land Property Document is added to the required document list.
AFI ₹1,00,000 or above	The conditional Land Property Document requirement is not triggered by this rule.

6. Document Checklist

Education: SC Certificate, Identity / Aadhaar Proof, Income Certificate, Admission / Course Proof and Bank Account Proof. Land Property Document is additionally required only for education-loan applications where AFI is below ₹1,00,000.

Business: SC Certificate, Identity / Aadhaar Proof, Income Certificate, Project Report and Bank Account Proof.

7. System Architecture

The browser-based frontend communicates with an Express REST API. The API handles account creation/sign-in, application records and document uploads. PostgreSQL can persist application and document metadata when DATABASE_URL is configured; otherwise, the prototype uses an in-memory fallback for these records. The local backend stores applicant documents beneath the Application directory.

Technology: HTML5, CSS3, JavaScript, Leaflet.js, OpenStreetMap, Google Maps direction URLs, Tesseract.js, PDF.js, jsQR, Node.js, Express.js, Multer and PostgreSQL via pg.

8. Typical User Workflow

Home → Login / Sign up → Dashboard → Search scheme → Apply for selected scheme → Enter applicant details → Analyze eligibility → Review partner locations and directions → Upload/capture required documents → Submit documents.

9. Setup and Execution

Install Node.js and npm. In a terminal, open the backend folder and run npm install, then npm start. The API listens on http://localhost:5000. For persistent database storage, create a PostgreSQL database, execute database/schema.sql and configure DATABASE_URL in a private backend/.env file. Open frontend/index.html for a basic preview; for camera and browser security features, use localhost or HTTPS rather than relying on file:// URLs.

10. Limitations, Privacy and Future Work

This prototype uses configured/demo scheme and partner data. Partner names, locations, availability/load values, scheme amounts, interest rates and eligibility conditions must be validated against current official sources. OCR, QR detection and Aadhaar checksum validation do not establish document authenticity or UIDAI identity authentication. Verifier endpoints are demonstration endpoints and require production-grade authentication and authorization.

Before production deployment, add secure role-based access, HTTPS, secret management, encrypted private document storage, malware scanning, audit logs, rate limiting, privacy/consent notices, retention and deletion controls, robust server-side validation, official scheme-data integrations and authorized identity/document verification.

Potential future improvements include verified live scheme feeds, secure applicant/verifier dashboards, status notifications, accessibility and localization improvements, more robust document classification, and integration with authorized partner directories.

11. Conclusion

NidhiRoute AI demonstrates an end-to-end concept for scheme discovery and application preparation. By combining searchable scheme information, scheme-specific forms, transparent prototype checks, EMI support, partner maps and document screening, it provides a coherent demonstration journey. Its next stage is to replace demo data and screening assumptions with verified official information, secure services and authorized integrations.

Project note: This report describes the included prototype files and their configured behavior. It is not an official scheme notification, financial offer or eligibility certificate.